

Phase Smoothing for Mutual Coupling Mitigation in Multifunctional Metasurfaces Designed with Diffractive Neural Networks

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Multifunctional metasurfaces hold great promise for next-generation communication systems, enabling spectral efficiency, dynamic adaptability, and hardware scalability for performing multiple wavefront manipulation tasks. However, traditional designs often neglect critical near-field interactions, leading to performance degradation in complex aperiodic configurations. In this work, an AI-driven design model is presented for a single-layer, passive, transmission-mode multifunctional metasurface capable of projecting different holographic patterns at 86 GHz (W-band) through spatial multiplexing of the input excitation. The approach integrates a diffractive neural network with an optimization algorithm and a phase-smoothing regularization scheme to mitigate mutual coupling between unit cells. The proposed model refines the phase profile to balance target performance and physical constraints, enhancing the validity of the local periodicity approximation. The experimental results closely match the simulations, validating the proposed method. This work demonstrates that an AI-driven model, enhanced by physical regularization, offers a scalable and efficient path toward high-performance multifunctional metasurfaces, enabling dynamic-like functionalities without active components—crucial for next-generation wireless communication systems.

to traditional microwave frequencies—enabling unprecedented capacity and ultra-high throughput^[1,2]—practical implementations remain constrained by the bulk, power inefficiency, and limited reconfigurability of conventional radio-frequency (RF) components. Metasurfaces, artificially engineered planar structures comprising subwavelength unit cells, offer a transformative solution by enabling precise electromagnetic (EM) wave manipulation, including phase, amplitude, and polarization control.^[3–5] These ultracompact devices achieve advanced functionalities such as reconfigurable beam steering,^[6,7] EM cloaking,^[8,9] dynamic holography,^[10,11] orbital angular momentum (OAM) multiplexing,^[12,13] or focusing.^[14,15]

Lately, metasurfaces have evolved toward active architectures to translate their ultracompact EM-wave manipulation capabilities into adaptive wireless platforms. Typically, reconfigurable intelligent

1. Introduction

The transition to next generation wireless systems demands spectral efficiency, dynamic adaptability, and hardware scalability. While millimeter-wave (mmWave) and terahertz (THz) bands provide significantly larger bandwidth compared

surfaces (RIS) rely on tunable elements (e.g., PIN diodes, MEMS, varactors, liquid crystals, or phase-change materials) to achieve dynamic control and real-time reconfiguration responses to support multiple functions.^[16–18] However, their integration faces critical challenges in the mmWave/THz bands, such as high insertion losses, limited availability of tunable materials, and

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fabrication complexity due to subwavelength unit cell dimensions. These limitations hinder their practical implementations in wireless systems operating in the mmWave band.^[13,19] These challenges motivate the exploration of passive, illumination-controlled alternatives that circumvent active component limitations.

In contrast to RIS, non-reconfigurable metasurfaces with multifunctional capabilities offer an appealing alternative in EM wavefront engineering.^[20,21] Unlike reconfigurable systems, these platforms shift dynamic control to illumination parameters, utilizing variations in the polarization of the incident wave, angle of incidence, or spatial multiplexing.^[22–24] By combining static phase profiles with tailored excitation conditions, they achieve multiplexed functionalities,^[25–28] but their design using traditional methodologies typically relies on laborious and time-consuming trial-and-error procedures, extensive parameter sweeps, and phase retrieval algorithms.^[29] To address these barriers, researchers are increasingly adopting artificial intelligence (AI) to automate design workflows and enhance device performance.^[30–34] Recent efforts leverage inverse design techniques^[35–38] and diffractive neural networks^[28,39–43] to accelerate metasurface optimization.

These AI-based models are grounded in theoretical EM formulations, such as the Huygens–Fresnel (HF),^[44] and Rayleigh–Sommerfeld^[39] principles. However, they rely on the local periodicity approximation, which is a simplification where unit cells are optimized as if they reside in an infinite periodic array. Real-world metasurfaces, however, require spatially varying phase profiles to achieve the intended performance. Abrupt phase transitions between adjacent cells violates this assumption, introducing unmodeled mutual coupling that distorts wavefronts and degrades performance. **While AI accelerates design workflows, most frameworks neglect mutual coupling effects, with a notable exception being studies that train deep neural networks (DNNs) to quantify actual EM responses.**^[45–47] However, these DNNs impose substantial computational overhead, as they require full-wave simulations or experimental data to account for near-field interactions, a limitation that hinders scalability.

Existing strategies to address mutual coupling fall into two distinct approaches: modeling frameworks and mitigation strategies. Modeling frameworks incorporate coupling effects into the design process through methods such as macroperiodicity-based architectures,^[48,49] analytical circuit models,^[50] numerical techniques like local phase analysis,^[51] and coupling matrices.^[52,53] These methods focus on predicting and compensating for coupling-induced deviations during design. Mitigation strategies, on the other hand, reduce coupling via structural or material innovations, such as elliptical unit cells,^[54] and Z-shaped elements,^[50] which inherently minimize near-field interactions. While these approaches improve accuracy or reduce coupling, they often impose trade-offs in structural design flexibility, computational efficiency, or bandwidth.

Our work shifts the focus toward integrating phase coherence and mutual coupling mitigation into a single-layer multifunctional metasurface for operation at mmWave frequencies. We present a transmission-mode metasurface operating at 86 GHz (W-band) that generates four distinct holographic images via spatial multiplexing across its quadrants. This passive architecture achieves versatile multi-output holography without active

components or complex switching mechanisms, leveraging illumination at different areas. Building on recent advances that combine inverse design with gradient descent and incorporate a phase-smoothing loss term to control abrupt phase transitions,^[38] our approach extends this concept in several key directions. In particular, we include experimental fabrication and validation, which is essential for assessing practical feasibility. Furthermore, our diffractive neural network adopts a different architecture, and we show that even a relatively simple regularization strategy—phase smoothing via Laplacian convolution—can effectively mitigate mutual coupling.

The proposed Adam-optimized neural network treats metasurface design as a constrained optimization problem, iteratively refining the phase profile to simultaneously minimize 1) the loss error between predicted and target holographic patterns, and 2) mutual coupling between adjacent unit cells. To address abrupt phase transitions that violate the local periodicity assumption, we introduce a phase-smoothing algorithm within the NN training loop. This algorithm enforces gradual phase gradients, preserving the validity of unit-cell approximations while suppressing coupling-induced wavefront distortions. Numerical simulations confirm that unsmoothed designs exhibit significant deviations from theoretical expectations due to mutual coupling, while smoothed profiles show marked improvement (specific performance metrics will be quantified in the section Analysis and Simulation). Experimental validation demonstrates a dielectric metasurface generating four distinct holograms depending on the illumination position of the emitting antenna, with measurements aligning closely with simulations (detailed analysis provided in the section Experimental Results).

2. Analysis and Simulation

Figure 1a illustrates the operational principle of our multifunctional metasurface. A horn antenna illuminates the metasurface at four positions (quadrants), generating four different holographic patterns on the observation plane. The metasurface, positioned between the input (antenna) and output (hologram) planes, encodes these functionalities through a spatially varying phase profile. Therefore, this static design eliminates the need for active components while enabling position-dependent reconfigurability.

The metasurface consists of a matrix of dielectric posts arranged on a lattice with a size of $l_1 = 1.40$ mm (**Figure 1b** left panel), where the height of each post is individually tailored to achieve the desired phase modulation,^[28,39,55–58] resulting in an aperiodic overall structure. Following the standard methodology for holographic metasurface design,^[59] the EM response of each post was characterized by simulating a periodic unit cell with identical cross-sectional dimensions using the Frequency Domain Solver of CST Studio Suite with Floquet boundary conditions under normal incidence and linear polarization. The posts are made of DM220 Therman Resin ($\epsilon_r = 3.1$, $\tan \delta = 0.02$) and feature a top square cavity ($l_2 = 1$ mm width, $t = 0.65$ mm depth) acting as a quarter-wavelength impedance transformer to minimize reflection losses^[39]. By varying the height h between 1.40 and 6.00 mm, we obtained a 0 – 2π phase coverage at the 86 GHz operating frequency, with a transmission magnitude above -2 dB (**Figure 1b** right panel).

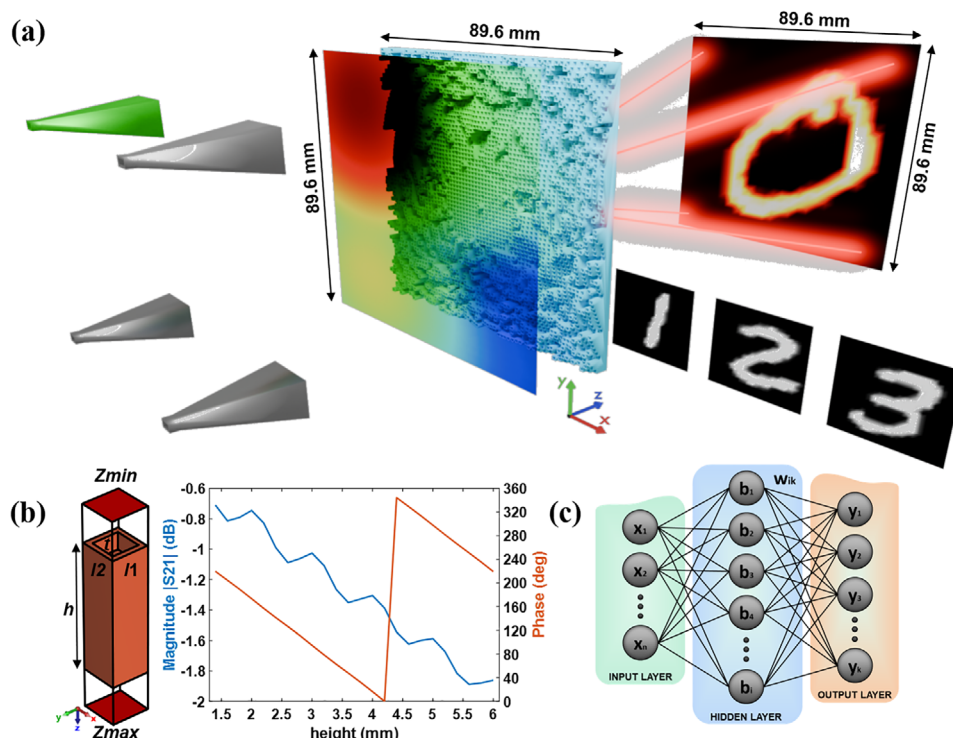


Figure 1. a) Schematic of our Diffractive Neural Network (DNN), showing the antenna field sources (input layer), the metasurface (hidden layer), and the target holographic patterns (output layer). The top-left green antenna is active to generate a specific hologram; other antennas produce the patterns of digits 1, 2, or 3. b) Unit cell schematic with dimensions and simulated transmission coefficient (magnitude and phase) as a function of post height at the operating frequency of 86 GHz. c) Conceptual node-based representation of the DNN in (a): input nodes ($x_1 \dots x_n$) correspond to the antenna field at each point on the metasurface and connect to hidden nodes ($b_1 \dots b_n$) through local phase modulation (biases). Hidden-to-output connections (w_{ik}) are computed via Huygens–Fresnel propagation to the image plane.

The architecture in Figure 1c is equivalent to Figure 1a, which operates as a diffractive neural network, establishing an analogy with conventional NNs. In this framework, the input layer corresponds to a normally incident field, the hidden layer is represented by the metasurface itself, and the output layer maps to the target amplitude patterns projected onto the observation plane.

First, a dataset of input-output combinations was constructed by varying the illumination position to excite different zones of the metasurface and generate multiple holographic patterns. Input fields were obtained by simulating the electric field distribu-

tion of a Standard Gain Horn antenna at various distances using CST Studio Suite. Several antenna-to-metasurface distances were evaluated to get a trade-off solution between amplitude uniformity and phase diversity. A quasi-uniform amplitude profile ensures effective illumination of each metasurface zone, while sufficient phase variation is crucial to enable distinct responses under different excitations, which is essential to achieve multifunctionality. An optimal trade-off was found at a 100 mm separation (Figure 2a,b). Target output patterns were selected from digits 0–3 of the MNIST handwritten digit dataset (Figure 2c).^[60]

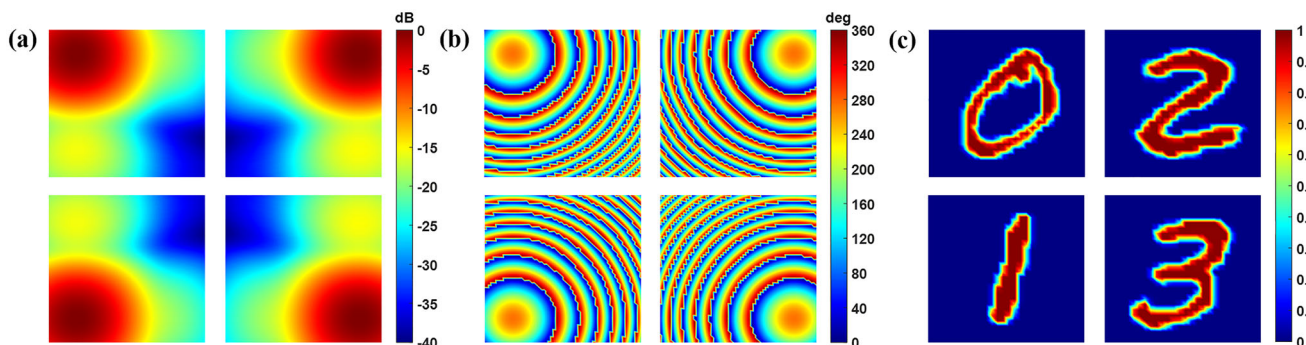


Figure 2. Simulated electric field of the horn antenna at 86 GHz and at a distance of 100 mm, showing a) magnitude and b) phase for four zones excitations. c) Amplitude of target patterns from the MNIST dataset.

In traditional NNs, interlayer connections are defined by trainable parameters (weights). In our framework, however, these connections are governed by wave propagation physics and remain fixed, derived directly from the HF diffraction principle. To model this, we discretized the input plane (64×64 source points), metasurface (64×64 unit cells), and output plane (28×28 pixels) into grids with the same dimensions ($89.6 \times 89.6 \text{ mm}^2$), treating each grid element as a neuron. The weights between the input field and the metasurface are unitary, to directly transfer the simulated antenna field (Figure 2a,b) to the metasurface input side. Here, each point on the metasurface acts as a secondary wave source, radiating contributions to every point on the observation plane. For a metasurface plane with coordinates (x_i, y_i, z_i) and an output plane with coordinates (x_k, y_k, z_k) , the weight w_{ik} between the i -th metasurface unit cell and the k -th output pixel is determined by the following equation:

$$w_{ik} = \frac{1}{2r_{ik}} \left(1 + \sqrt{1 - \left(\frac{y_i}{r_{ik}} \right)^2} \right) \exp^{-jk_0 r_{ik}} \quad (1)$$

where $r_{ik} = \sqrt{(x_i - x_k)^2 + (y_i - y_k)^2 + (z_i - z_k)^2}$ is the Euclidean distance between the i -th metasurface unit cell and the k -th output point; and k_0 is the free-space wavenumber. This formulation inherently embeds the wave physics within the network architecture, replacing the need for trainable weights.

To optimize the phase profile we employed backpropagation frameworks,^[61,62] training the NN using the Adam gradient-based optimization algorithm.^[63] The mean square error (MSE) was used as the loss function (Equation 2) to evaluate the model's efficiency and quantify the discrepancy between the desired and obtained output patterns,

$$L_{\text{MSE}} = \frac{1}{N} \sum_{i=1}^N (E_{\text{patt}} - E_{\text{pred}})^2 \quad (2)$$

where E_{patt} and E_{pred} denote the target and predicted amplitudes, respectively, and N is the number of output points. Optimization is achieved by varying the height of each unit cell, which directly controls the local phase shift (ϕ). This geometric parameter functions analogously to the bias term in conventional NNs. By tuning the height of each unit cell, we effectively bias its EM response, enabling control over the output wavefront.

After 30 000 epochs with a fixed learning rate of 0.01, the model converged, with a minimum MSE value of 0.0119, yielding an optimized phase profile across the metasurface (Figure 3a). The minimal discrepancy and high precision between the original MNIST digits (Figure 2c) and the generated holographic counterparts (Figure 3c) demonstrate the accuracy of our neural network implementation.

To validate our findings, full-wave numerical simulations of the complete structure were carried out using the CST Studio Suite time-domain solver. The illuminating antenna field at a distance of 100 mm, was positioned on the input side of the metasurface, sequentially illuminating each region. An electric field monitor captured the field throughout the volume, and a cut was taken on the output plane located 100 mm from the metasurface at 86 GHz. As illustrated in Figure 3e, the simulated field dis-

tribution is significantly different from that obtained in the HF analysis, and fails to reproduce the intended digit features, rendering them unrecognizable, with a MSE value of 0.1354.

To confirm that the discrepancy arises from physical interactions within the metasurface rather than from differences in field computation methods, we imported the analytically calculated field distribution—obtained by multiplying the incident field with the metasurface response—into CST and simulated it directly. This study is presented in the Section 1 of the Supporting Information, where we analyze a scenario involving nine holographic patterns generated by nine point-source illuminations on a metasurface of the same size. In this case, the patterns were accurately reproduced, indicating that the mismatch observed in full-wave simulations originates from mutual coupling effects between neighboring elements, which are not captured in the initial analytical model or in the idealized simulation. Beyond that main target, the results serve to highlight the method's scalability, showing its capacity to support a higher number of functionalities within a single metasurface.

This mutual coupling is a physical phenomenon stemming from near-field interactions between neighboring elements, and it is not accounted for by the local periodicity approximation used in the design methodology. This approximation treats each unit cell as part of an infinite, uniformly periodic array and neglects the influence of neighboring elements with differing geometries. This is in contradiction with the complex phase distribution required for multifunctional operation, which leads to substantial height variations between adjacent posts (Figure 3a), enhancing near-field interactions that invalidate the local periodicity assumption. As a result, the isolated unit cell simulations used to construct the HF model fail to capture the true electromagnetic behavior of the metasurface, ultimately degrading the quality of the projected holograms.

To address this problem, we developed the custom loss function in equation 3, which incorporates a novel gradient-based regularization term to promote phase smoothness. In addition to the standard MSE between calculated and desired field amplitudes (weighted by a fitting factor $1 - s$, where $s \in [0, 1]$ is the smoothing value), this new term operates on the phase distribution of the holographic plate ϕ , reshaped into a 2D matrix. By applying convolutional filters K_n , (where n is odd, ranging from 1 to $u - 1$, incrementing by two, and u is the number of unit cells) to compute local phase gradients $L_{\text{grad}}^{(n)}$, normalizing these gradients, and calculating their mean, we penalize abrupt phase variations. The regularization term is added to the total loss, weighted by the smoothness s and a factor dependent on the number of filters used $N = \frac{u}{2}$. The model, controlled by the smoothness parameter, aims to generate phase profiles with more gradual transitions, thereby reducing the impact of mutual coupling and improving the performance of the multifunctional design. The complete formulation is presented in the following equations:

$$L_{\text{total}} = (1 - s) \cdot L_{\text{MSE}} + \frac{s}{N} \sum_{\substack{n=1 \\ n \equiv 1 \pmod{2}}}^u L_{\text{grad}}^{(n)} \quad (3)$$

$$L_{\text{grad}}^{(n)} = \frac{1}{H_n W_n} \sum_{i=1}^{H_n} \sum_{j=1}^{W_n} \left| (\phi * K_n)_{i,j} \right| / \max(|\phi * K_n|) \quad (4)$$

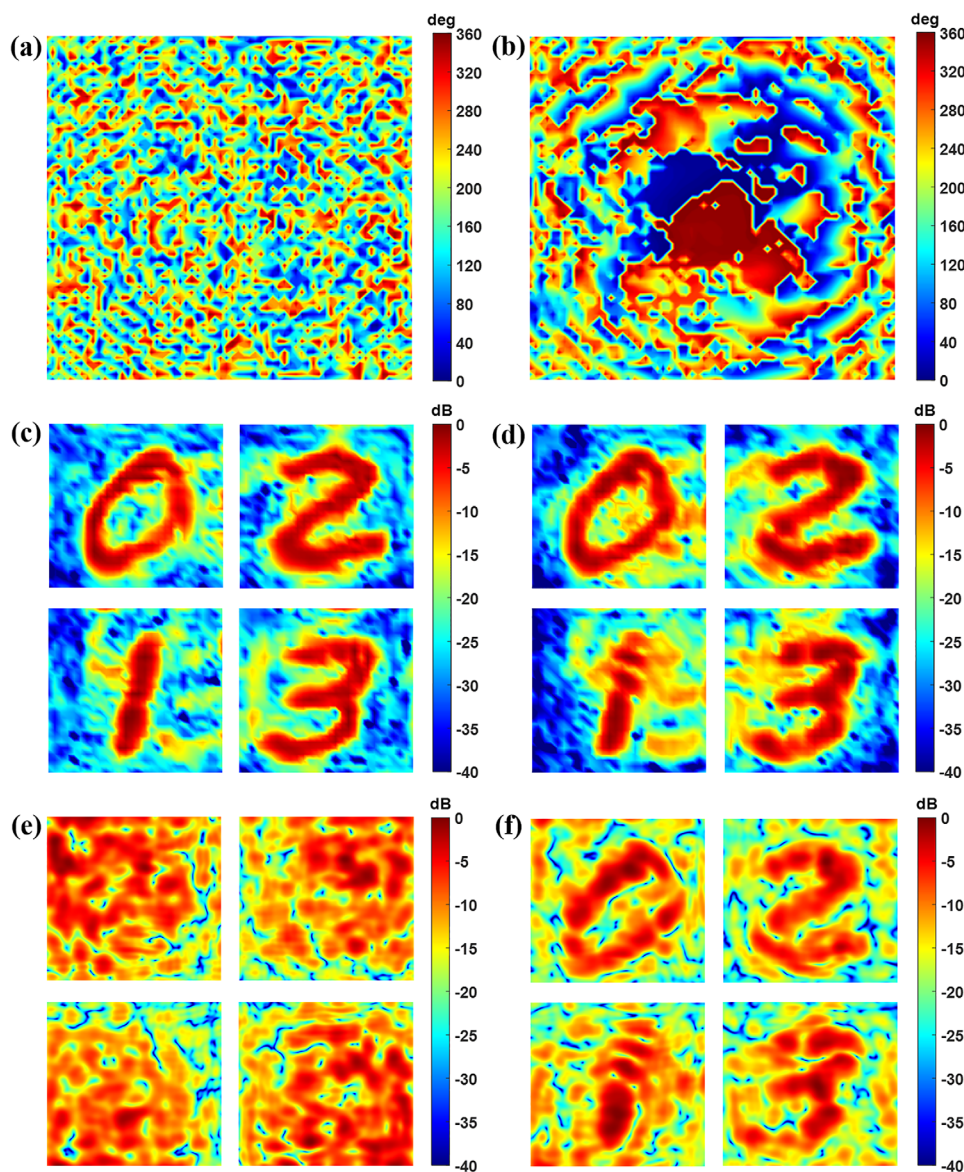


Figure 3. Comparison of phase profiles and pattern amplitudes. a) Optimized phase profile (no smoothing), b) Optimized phase profile (smoothing factor 0.7), c) Predicted pattern amplitudes (no smoothing), d) Predicted pattern amplitudes (smoothing factor 0.7), e) Simulated patterns (no smoothing), and f) Simulated patterns (smoothing factor 0.7).

where H_n and W_n are the dimensions of the resulting feature map, and $*$ denotes the valid convolution operation. K_n is commonly known as the discrete Laplacian kernel and is typically used for edge detection.

We refined the metasurface phase distribution with the new loss function. After several tests, different smoothing values and 30 000 epochs, **Figure 4** show the MSE performance of the NN.

An expanded analysis of this parameter indicates that smoothing values near 0 yield lower MSE values but lead to strong phase variations, effectively reverting to the initial design which deviated significantly from the local periodicity approximation. Conversely, values approaching 1 prioritize phase smoothness over output precision, but degrade the generated hologram due to an increased MSE. In fact, when $s = 1$, the model focuses entirely

on minimizing plate gradients, and drives the MSE to zero (trivial solution) without producing any meaningful holographic pattern. This indicates that a lower MSE value does not necessarily imply better image quality, as it may correspond to an overly smooth but non-functional design. To illustrate this trade-off, **Figure 5a–g** shows how varying s from 0 to 1 impacts both the output plane and the metasurface phase profile. As s increases, the output patterns degrade due to increasing MSE (L_{MSE} from 0.0119 to 0.1604), while the metasurface becomes progressively smoother (L_{grad} from 0.0322 to 0.0157).

Notably, the error curves for smoothing values between 0 and 0.7 exhibit limited variation, indicating a similar performance in this range. Therefore, we choose $s = 0.7$ as an optimal compromise, offering a smooth phase distribution while maintaining

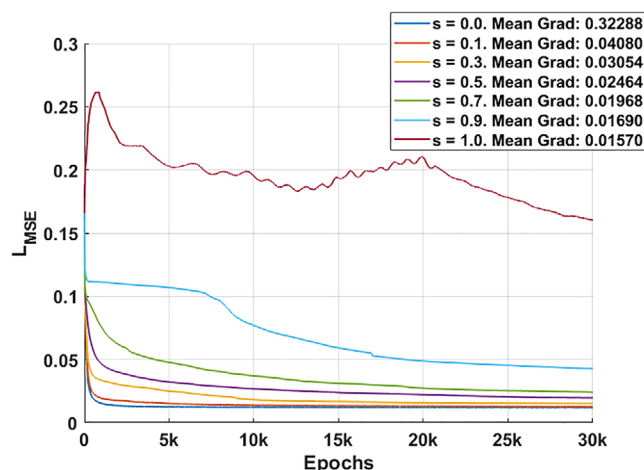


Figure 4. Evolution of the mean squared error (MSE) and the corresponding mean gradient as functions of the smoothness parameter $s \in [0, 1]$, evaluated over 30 000 training epochs.

an MSE comparable to other values in the clustered region. As shown in Figure 3b, the resulting phase profile exhibits a considerably smoother transition compared to the previous design. Although this leads to slightly degraded results in the analytical HF model—see Figure 3d vs Figure 3c, with an increased MSE from 0.0119 to 0.0242—the new structure better satisfies the local periodicity approximation. This improvement is crucial, as it enables the generation of high-quality holograms in the full-wave numerical simulations, shown in Figure 3f, improving the MSE value from 0.1354 to 0.0708.

The simulated holograms accurately reproduce the target patterns and show a marked quality enhancement over the original, unsmoothed design. To assess the robustness of this approach, we also simulated a metasurface designed with a smoothing value of $s = 0.5$ (not shown). Although this lower smoothing yields a slightly higher MSE (0.0719) than the $s = 0.7$ case, it nonetheless produces acceptable holograms while tolerating a modest deviation from the local-periodicity approximation. These results demonstrate that the smoothing parameter can be tuned to balance fidelity to the local-periodicity approximation and hologram quality according to specific design requirements.

Although now the simulated results are in good agreement with the theoretical ones, there are small discrepancies that are explained by two main factors. First, the HF equation is an approximation that does not consider crosscoupling between unit cells. Second, this mutual coupling between metasurface elements, although mitigated by the new loss function, cannot be completely eliminated. These residual interactions between cells slightly affect the field distribution and consequently introduce small deviations from the theoretical model. Despite these limitations, the results validate the effectiveness of the approach employed and highlight the importance of considering the effects of mutual coupling to improve the accuracy of the multifunctional design. The number of holographic patterns that can be achieved in our design depends on several factors: the number of unit cells (i.e., the metasurface size), the illumination conditions, and the distance to the observation plane. In fact, combining multiple

input beams also increases the number of achievable output patterns.

3. Experimental Results

The metasurface (Figure 6b) was fabricated using a high-resolution DWS-020X 3D printer, chosen for its ability to accurately reproduce the sub-millimeter features of the dielectric unit cells. This additive manufacturing technique ensured precise realization of the designed geometries, including the critical cavity on top. We characterized the metasurface using the experimental setup shown in Figure 6a. A Flann Microwave Ltd. Standard Gain Horn Antenna (Model 27240-20) was used to illuminate the metasurface, while a WR10 near-field probe scanned an output plane of square size ($100 \times 100 \text{ mm}^2$) in 1 mm steps. To ensure accuracy, the metasurface was carefully centered between the antenna and probe, with all components aligned to reduce positioning errors and maintain measurement consistency.

The experimental results, shown in Figure 6c, strongly support the theoretical predictions. The generated holograms reproduce the target patterns, as evidenced by the low overall MSE value 0.102. This finding serves as the experimental demonstration of the importance of incorporating a phase smoothing parameter during the optimization process, which minimizes abrupt phase transitions between adjacent elements and thereby ensures optimal metasurface performance. Moreover, the proposed design effectively generates multiple outputs from multiple inputs in a completely passive setup, fully in agreement with the predictions made by the NN model. Finally, the generated holograms exhibit lower fidelity compared to optical implementations,^[20,21,64,65] as is typically the case at mmWave frequencies. The physical origin of this difference, related to aperture scaling and illumination constraints, is discussed in Section S2 (Supporting Information).

To provide a quantitative assessment of the metasurface's operational bandwidth within the 72 – 100 GHz range, Figure 7 presents the MSE between the analytical targets and the reconstructed patterns for both simulation (top row) and measurement (bottom row), using two analytical models: an idealized model (left column), assuming frequency-independent unit-cell response (propagation effects only), and a realistic model (right column), including the simulated frequency response of each unit cell (propagation + dispersion), see Section S3 (Supporting Information). In all analyzed cases, the MSE decreases with increasing frequency, with average values of 0.0245 for simulations and 0.0411 for measurements. Interestingly, the trends and absolute values are very similar for the idealized and realistic models, indicating that the metasurface's broadband performance is robust and only weakly affected by unit-cell dispersion in this frequency range. This agrees with the simulated transmission coefficient maps shown in Figure S4 (Supporting Information), where the magnitude depends mainly on height with little dependency on the frequency, while the phase depends on both parameters but with phase differences between unit-cell heights that remain nearly constant across the band. The poorer performance observed at lower frequencies, particularly evident in the measurements, can be attributed to the larger wavelength reducing the effective aperture size in terms of wavelengths, which limits the achievable angular resolution and amplifies mutual coupling effects between neighboring unit cells, thereby degrading

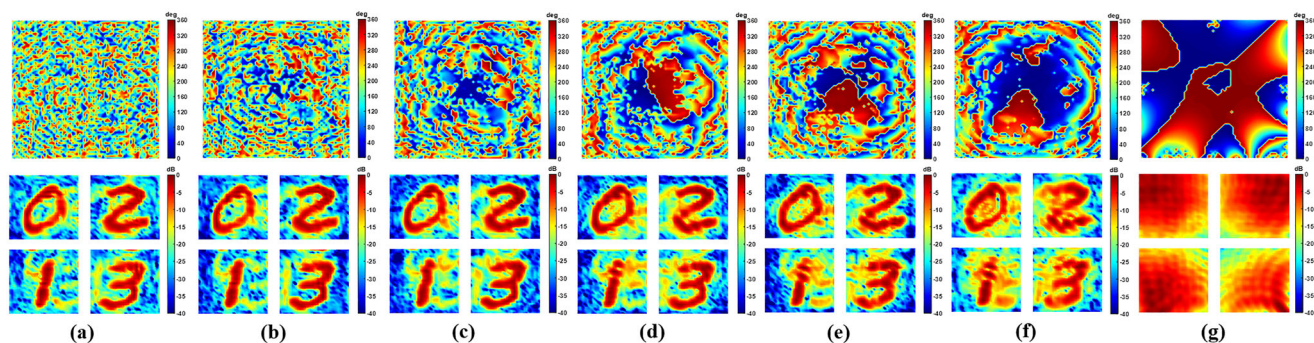


Figure 5. Evolution of the optimized phase profiles (top row) and the corresponding magnitude target patterns (bottom row), obtained under the Huygens-Fresnel approximation, as the smoothness parameter s increases. Results are shown for a) $s = 0$, b) $s = 0.1$, c) $s = 0.3$, d) $s = 0.5$, e) $s = 0.7$, f) $s = 0.9$, and g) $s = 1$.

the fidelity of the reconstructed holograms. All these observations are consistent with the visual inspection of the holograms in Figures S5–S8 (Supporting Information) and the evolution of the measured holograms shown in Videos S9a–S9d (Supporting Information).

A closer inspection of the MSE by target digit reveals that, in simulations, the largest errors occur for digits “1” and “0,” whereas in measurements they occur for “3” and “0.” In the experimental case, several factors contributed to these differences: slight misalignments between the horn antenna aperture and the

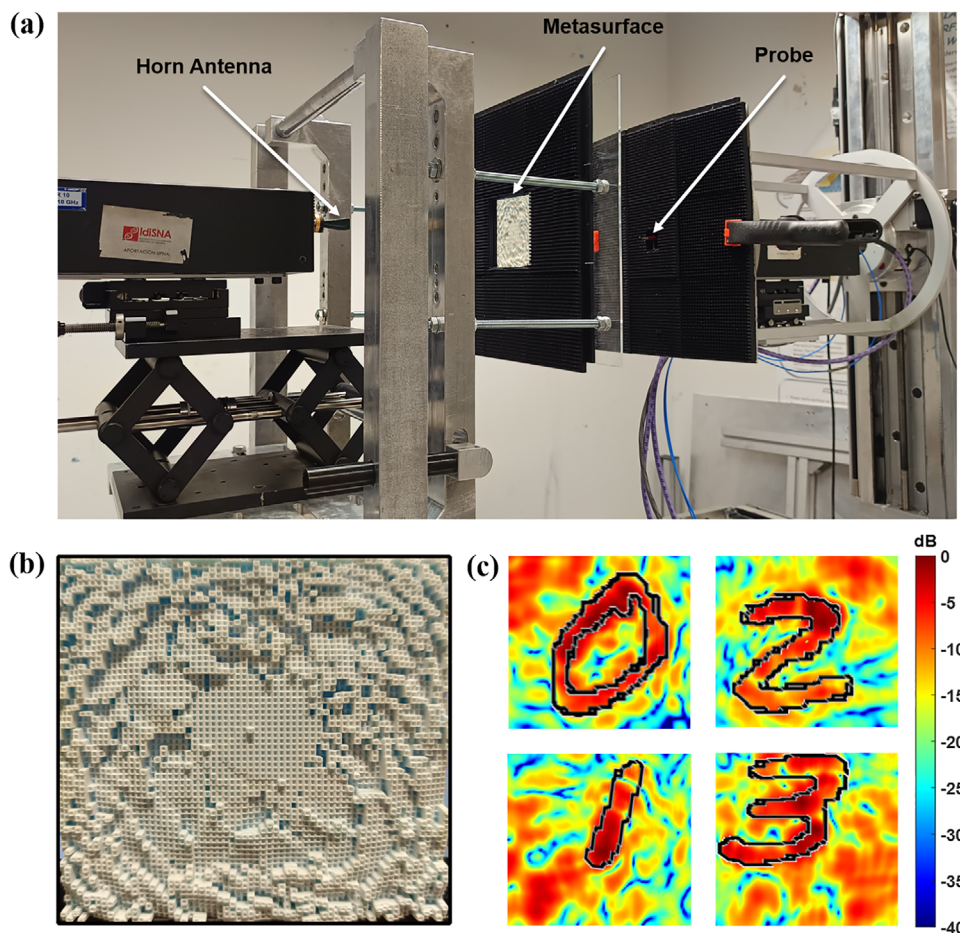


Figure 6. a) Experimental setup, including the illuminating antenna, metasurface, and near-field scanning probe. b) Image of the fabricated dielectric metasurface. c) Measured holographic patterns at the output plane for the four illumination configurations. To visually corroborate the results, a border outlining the theoretical digit shape has been added to each pattern.

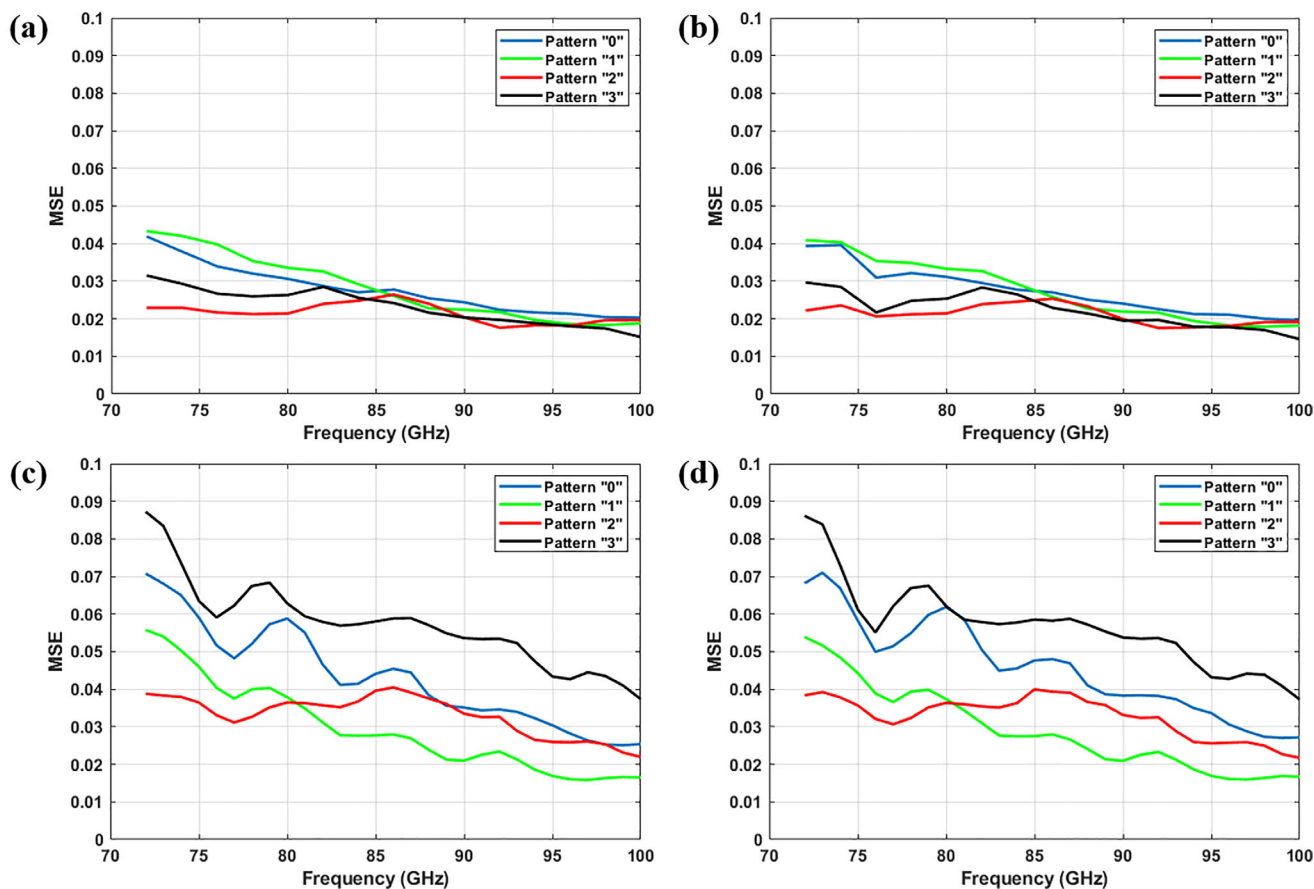


Figure 7. MSE as a function of frequency for each digit (0–3). a) MSE between the analytical target and simulated patterns (idealized model). b) MSE between the analytical target and simulated patterns (realistic model). c) MSE between the analytical target and measured patterns (idealized model). d) MSE between the analytical target and measured patterns (realistic model).

metasurface, leading to small lateral shifts in certain digits (notably “1” and “3”); power artifacts in the plane corners linked to the relative horn position during each projection; and variations in the dielectric properties of the substrate combined with fabrication tolerances, which introduced additional phase errors. Together, these effects tend to degrade more strongly the fine features of some digits (e.g., “3”) and explain the slightly higher MSE in measurements.

Finally, the decrease in MSE at higher frequencies is due to the fact that the MSE is computed against a frequency-dependent analytical target (idealized or realistic) that already incorporates the expected dispersion due to propagation and, in the realistic case, the frequency-dependent transmission. Consequently, a lower MSE at higher frequencies indicates a closer match to this reference; it does not necessarily imply an improvement in the absolute image quality, but rather a better agreement with the frequency-appropriate target.

4. Conclusion

We have developed a NN model integrating an optimization algorithm and gradient-based regularization to design smoothed phase profiles for a single-layer, transmission-mode multifunctional holographic metasurface operating at 86 GHz (W-band).

This approach enables precise wavefront control, effectively demonstrating a multiple-input multiple-output (MIMO)-like functionality through spatial multiplexing of the input excitation. The model accurately predicts the optimal phase distribution for each unit cell, allowing the projection of four distinct holograms onto an observation plane 100 mm from the metasurface by simply repositioning a standard gain horn antenna. The strong agreement between simulation and experimental results validates the effectiveness of our AI-driven design methodology in realizing complex holographic functionalities at mmWave frequencies. A key contribution of this work is the explicit consideration of reducing mutual coupling effects between unit cells through the implementation of a phase smoothing algorithm. By enforcing gradual phase transitions, we reduce deviations from the local periodicity approximation, leading to improved overall performance and accuracy of the holographic projections, a critical factor often overlooked in traditional design approaches for complex aperiodic metasurfaces. Importantly, the metasurface also demonstrates robust broadband performance across the 72 – 100 GHz range, indicating that the optimized phase profiles maintain high fidelity even under frequency variations, a crucial aspect for practical mmWave and 6G applications. This study highlights the significant potential of diffractive neural networks as a powerful and efficient tool for the design of

Table 1. Neural Network Summary.

Layer (type)	Output Shape	Param #
Source (InputLayer)	(4096)	0
Metasurface (HiddenLayer)	(4096)	4096
Pattern (OutputLayer)	(784)	0
Total params: 4096		
Trainable params: 4096		
Non-Trainable params: 0		

advanced, high-performance multifunctional metasurfaces operating in the challenging mmWave band, offering a pathway toward realizing dynamic-like functionalities without relying on complex active components. The proposed methodology provides a fast, reliable, and scalable solution for a range of emerging 6G applications requiring high-resolution holographic projection, secure wavefront shaping, and efficient short-range information transmission, with significant implications for next-generation communication, advanced imaging, and high-frequency sensing technologies.

5. Experimental Section

NN Model Design: The NN model was developed in Python using the TensorFlow library. Layers and loss function have been customized to perform the transformations from input fields to the desired output patterns, involving the manipulation of complex fields. The antenna field at a distance of 100 mm is discretized into 64×64 points. To this field is added the phase contribution of each unit cell (64×64), and multiplied by the complex weight of each connection with the output layer, which is discretized in 28×28 pixels. These weights are calculated with the expression (1) derived from the HF principle. The output must correspond to the target patterns located 100 mm from the metasurface. The Table 1 provides a summary of our model implementation.

Fabrication: The metasurface was fabricated using a high-resolution DWS Systems stereolithography 3D printer (Model 020X) in combination with DM220 Therra resin, a ceramic-loaded photopolymer optimized for high thermal stability and dielectric performance. This additive manufacturing approach allowed for the precise reproduction of the designed sub-millimeter features, including the complex cavity structures on top of each unit cell. The use of Therra resin ensured dimensional accuracy during and after curing, while also providing the necessary requirements for holographic multifunctional metasurface behavior. Based on our measurements, the fabrication process introduced a dimensional deviation of ± 0.15 mm, which was considered the fabrication error margin across the metasurface structure.

Supporting Information

Supporting Information is available from the Wiley Online Library or from the author.

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Conflict of Interest

The authors declare no conflict of interest.

Author Contributions

M.B. and A.M. conceived the idea for this work. D.R. and M.A. conducted the neural network analysis and design. D.R. performed hologram generations and numerical simulations. D.R. and A.T. carried out the fabrication process. D.R. and J.T. realized the experimental measurements. D.R., A.T. and M.B. prepared the manuscript. A.T. and M.B supervised the project.

Data Availability Statement

The data that support the findings of this study are available from the corresponding author upon reasonable request.

Keywords

diffraction neural networks, holography, mmWaves, multifunctional metasurfaces, mutual coupling

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